1. HOUDINI: HOlistic approach to UnderstanD and target the metastatIc NIche

General information

1.a Applicants

*Name**	*Institution*	*Department*	*Email*	*Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)*	*For Erasmus MC applicants: clinician/ researcher/ mix*	*Role in Flagship [Main Lead/Lead/ Applicant]*
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Corresponding Lead: Wytske van Weerden

Planned start date of seed funding: *1-11-2022*

Primary Convergence theme: *Fundamentals of Health & Disease*

Secondary Convergence theme (*if applicable*): Choose an item.

Tertiary Convergence theme (*if applicable*): Choose an item.

Quaternary Convergence theme (*if applicable*): Choose an item.

Keywords: *Cancer metastasis - Therapy - Biomechanics - microfluidics - organ-on-a-chip - Ethics - Public Health*

1.b Summary (300/300 words)

Cancer is a leading cause of death worldwide, accounting for nearly one in six deaths. A stunning 90% of current cancer-related deaths is caused by metastatic disease. Metastatic cancer strongly impacts patient's quality of life. Besides the personal burden, metastatic cancer also has major societal and economic consequences. Clearly, we urgently need better and affordable treatments. We suggest a game-changing **holistic research approach** bringing in patient and citizen expectations to fuel and drive fundamental research from the very start. It is our ambition to establish novel prediction tools and treatment approaches to conquer cancer metastasis. While current therapies are targeting the tumour cells or the vascularisation or the immune system, it is **HOUDINI's (HOlistic approach to UnderstanD and target the metastatIc NIche) mission to understand the complex metastatic process, in order to simultaneously target not one, but multiple key players of metastatic disease**. To decipher microenvironmental cues of metastatic disease, we will engineer *in vitro* models that mimic the composition and architecture of human organs. For this, bio-engineers, cell biologists and clinicians will integrate advanced (single)cell and 3D tissue imaging with molecular and cell biology technologies. Converging with experts in ethics and health economy will connect societal and patient expectations, engagement with stakeholders to include early health technology assessment, and ethical guideline definitions. This will help to assure that our fundamental new insights and technological innovations will be acceptable and meaningful to guide clinical implementation. HOUDINI will further impact the biomedical research field by actively training and educating young researchers in the holistic principles of convergent research. While HOUDINI will initially focus on two of the most common sites of metastasis, bone and liver, extension to other organs will be actively pursued. As HOUDINI initiates a paradigm shift towards holistic fundamental research, it will alter the landscape of biomedical science.

1.c Public Summary (267/300 words)

Cancer is one of the leading causes of death. The spread of tumour cells to distant organs (metastasis) is difficult to treat and accounts for up to 90% of all cancer-related deaths. The quality of life of cancer patients is strongly compromised, yet patients' needs and wishes are usually not included in fundamental research. Here, we wish to change this limitation and put **the patient at the centre of our research program, HOUDINI**. It is our ambition to establish novel prediction tools and treatment approaches to conquer cancer metastasis. To ensure responsible innovation, we will engage with patients, citizen and other stakeholders. To develop successful treatments, we need to understand why tumour cells are attracted by specific tissues and how they manage to settle, grow and manipulate their new environment. The preferential metastatic growth of several cancer types in specific organs, such as liver and bone, suggest that specific organ microenvironments promote cancer cell invasion and colonization. The HOUDINI consortium is dedicated to investigate the metastatic microenvironment and its interactions with tumour cells. This requires model systems that reliably mimic the physiology of human tissues during metastasis. Recent advances in biomedical technologies allow us to create "mini-tissues" that reflect the biological complexity, architecture and mechanical characteristics of healthy (human) organs. Therefore, specialists with medical, biological, ethical, economic and technological expertise will unite to create biological meaningful models and novel assays to identify new prediction tools and therapeutic targets that specifically suppress metastatic progression. HOUDINI will further actively train and educate young researchers in the holistic principles of convergent research thereby initiating the transition towards society-guided biomedical research.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2283/2500 words for 2.a to 2.e

• The Flagship program contributes to the Convergence Health & Technology mission (1)
• The Flagship program objectives fit with the guidance per theme provided by the theme leads, and/or fit with more than one theme of Convergence Health & Technology (2)
• The Flagship program merges approaches and insights from distinct disciplines to reach its goals (3)
• The staff granted by this call actively brings together different disciplines within the program, and creates visibility and attracts talents, partners, and funding for the program (4)
• The Flagship program has a clear view on long term sustainability after the 5-year central funding period (5)
• The envisioned results of the Flagship program have the potential for a significant contribution to healthcare, society and/or have scientific impact which has the potential for long-term societal impact (6)
• The Flagship program provides opportunities for BSc/BA and/or MSc/MA students of the participating institutes to participate in (interdisciplinary) projects (7)
• The proposal defines the pathway of how this contribution can be realized. The pathway includes concrete research and/or impact pathway (societal and/or economical) valorisation strategies (8)

2.a Contribution to H&T mission and fit with H&T theme(s) (284 words)

The recent epidemiological transition from cardiovascular disease to cancer as the leading cause of death in developed countries, poses a huge challenge for global healthcare systems. To address this growing and urgent medical need, new technological and biomedical breakthroughs are needed to **predict and treat metastasizing cancer and foster lifelong health**. The mission of the HOUDINI program is **to identify the role of the tumour microenvironment in metastasis as a basis for developing novel prediction tools and treatment approaches** to conquer cancer metastasis. While current cancer research is strongly focussed on targeting cancer cells, we instead propose a holistic approach that also includes the microenvironmental factors that determine invasion and growth of cancer cells in healthy organs. This unique perspective requires convergence of scientific expertise, know-how and methodologies from medicine, biology, chemistry, physics, engineering, health management, ethics and philosophy. Our holistic approach perfectly aligns with the Convergence Health & Technology mission to improve long-term health, in particular through **Theme 1. Fundamentals of Health and disease** and **Theme 3. Human centred technology**. The HOUDINI consortium offers an interactive platform of engineers and scientists from the biomedical and social sciences and humanities domains that allows cross-fertilization between fundamental biomedical research and health technology development in connection with ethical and socio-economic research ensuring responsible development. Houdini's research endeavours will be further supported by an education and training program for students from biomedical, technical, social and management disciplines across the Erasmus MC, TUD and EUR. This will foster a new way of educating next generations of young transdisciplinary research professionals. Through its research and training activities, the HOUDINI network will attract other research groups and industrial partners to achieve innovations in research, education, health management, and societal awareness.

2.b Synergy between disciplines (326 words)

HOUDINI is a concerted effort of disciplines across EMC, TUD and EUR to serve its holistic mission. Innovative biomedical and technological developments aiming to establish **new prediction tools and therapies for disseminated disease, will be developed in line with patient's needs, financial acceptability and feasibility in clinical practice**. The biomedical and technological expertise of EMC and TUD partners to understand the processes in metastatic disease, will be connected to proficiencies of partners from EUR and EMC to define the associated economical and societal considerations. Furthermore, the development of innovative and dedicated human-based *in vitro* culture

systems calls for ethical considerations and patients' willingness to provide biomaterials and clinical data, leaning on the expertise of our EMC and EUR partners. **HOUDINI involves five interconnected PhD projects that encompass the entire range of disciplines.** New societal, biological and technological findings will be shared in recurrent HOUDINI meetings to guide research and development. The strong collaboration between the research projects will be further strengthened by providing parallel educational sessions to enhance mutual understanding across the various disciplines. To further extend the transdisciplinary training of students at all levels (BSc, MSc and PhD), HOUDINI will develop courses and workshops that will be integrated in existing programs such as the Nanobiology curriculum, the MA program Philosophy Now and the PhD Courses "Philosophy of Responsible Innovation" and “Stem cells, Organoids and Regenerative medicine” at the three different institutes. Collectively, this will yield **a new generation of scientists who are able to operate beyond their own expertise by integrating different angles and disciplines**, which will stimulate new original project ideas and research funding. HOUDINI will actively promote patient involvement (support letters from the “Nederlandse Leverpatiënten Vereniging” and “ProstaatkankerStichting”) and public awareness to help shape potential future applications generated within the program.
Finally, new insights and achievements will be actively disseminated to the scientific community and the general public to attract additional disciplines and stakeholders not yet involved in HOUDINI and enhance public awareness.

2.c Long term sustainability plan (571 words)

HOUDINI is aiming to create state-of-the-art, innovative scientific and technological research and training environment striving to understand and target metastatic cancer. To this end, HOUDINI has defined **five (PhD) projects, divided over three work packages,** based on the current expertise and knowledge of its partners that are deemed essential to build the program (Table 1 Project aims). Four biotechnological projects address mechanistic aspects of the metastatic cascade (WP 1 and 2), and one project is dedicated to integrate the societal aspects of this fundamental research within HOUDINI (WP 3), together being essential for the holistic approach. These projects will fill our **knowledge 'toolbox'** and provide dedicated models and read-outs that will likely support and be incorporated in future project applications. The PhD projects will be complemented with MSc-driven **pilot projects** focussing on dedicated, smaller tasks (generally 10-12 months) to generate proof-of-concept data, validate new hypotheses or test feasibility of potential new applications, that will generate preliminary data for new project applications. We plan to invoke a special **HOUDINI fund for innovative ideas** to get preliminary work with high potential to gain external funding of HOUDINI for year 3 and beyond. The current external matching of HOUDINI is guaranteed by the participating partners and based on their active line of research that aligns with the HOUDINI core project's tasks. **Most partners have received or applied for grant applications to warrant future external funding** (Table 2 and budget). Moreover, relevant partners, such as hDMT and BI/OND (section 4C and support letters) have shown their interest in supporting future collaborations. Finally, each core project has a dedicated task to provide new project funding based on the initial achievements (knowledge toolbox) and extend the scope of HOUDINI continuously.

The pilot projects of HOUDINI will assess new research ideas and applications of developed technologies, thus showcasing HOUDINI to interested parties to invoke new research collaborations **attracting external funding**. For **sustainability** of the program, in the first 2 years of the program we will stimulate and support PI-initiated grant application initiatives through providing grant writing support. Assuming a general success rate of 25% and a matching percentage of 20% from the project budget, this will provide the financial needs of HOUDINI beyond year 3. For continuous growth of HOUDINI beyond the first 5 years, we intend in year 3-4 of the program to actively establish collaborations with other (inter)national parties to apply for consortium-wide research and/or networks. The HOUDINI management (WP4) will provide grant writing support to optimally facilitate success in future applications. Moreover, HOUDINI, while building its community, will also actively seek new opportunities to attract collaborators from academia as well as partners from (non)profit enterprises. The close collaboration between the academic hospital, biomedical researchers, engineers and societal researchers is well-aligned with global trends to address societal challenges by transdisciplinary research in funding schemes such as NWA, KIC and 'Perspectief' consortia in the Netherlands and EIC Pathfinder calls in Europe. HOUDINI will actively explore and support such new initiatives from the start and also beyond the 5-year funding period.

Figure 1: HOUDINI scope development over time, starting from research focused on liver and bone metastasis in the first 5 years (centre) and expanding to other organs and readouts past year 5 (going outwards), with the ultimate aim to establish metastasis prediction tools and perform holistic therapy studies (outer edge).; Abbreviations: WP: work package; OoC: organ-on-chip; AFM/FFM: Atomic/Fluid Force microscopy; sc: single cell; comp: computational; spec: spectrometry.

2.d Visibility, attracting talents and new partners (607 words)

Pillar 1: attracting partners and funding. HOUDINI will converge biomedical research, bioengineering, education and societal engagement to serve one of the most urgent unmet medical needs of today: curing metastatic disease and maintaining optimal quality of life for the aging population. Various private stakeholders and patient organizations (BI/OND, “Nederlandse Leverpatienten Vereniging”, “ProstaatkankerStichting”) have already expressed their interest and support of our flagship. We start with liver and bone metastasis, but will broaden our scope to other cancer types in the years to come. Integrating universal technological and socio-economic studies of metastases enables expansion to the entire oncology realm. We plan to reach out to other clinical partners and related patient organizations already at an early stage of the program through peer-reviewed publications, and engage in conferences and educational programs to attract talented researchers.

Pillar 2: engaging students and early career researchers in interdisciplinary projects. We will train the HOUDINI PhD students in a transdisciplinary mind-set by providing secondments to other projects/project groups. We will also establish a series of seminars on translational topics connecting technology, biology, medicine, health economics, ethics and society presented by HOUDINI PIs, and open to interested young researchers and students from outside HOUDINI to engage them and attract talented students to work in this field. Moreover, we aim to organize hands-on technology workshops to enhance the application of HOUDINI's toolbox technology beyond its research field. Finally, to foster cohesion within the consortium, we will also establish an Early Career HOUDINI (Young HOUDINI) Network. To attract undergraduates, we will create internship positions that are jointly supervised by PIs from the three institutions for (BSc and MSc) students from EUR, EMC, LUMC, TUD and Universities of Applied Sciences (e.g., Biomedical Laboratory Science/BML at Hogeschool Rotterdam, Breda and Leiden). Education and training of young professionals can be integrated in existing educational programs such as 2nd year programs (*'Regenerative Medicine for Doctors of Tomorrow')* and Minors for 3th year bachelors (Regenerative Medicine; coordinated by Verstegen, vd Eerden, Bunnik, Fratila-Apachitei) and Graduate School *Biomedical Science Track PhD course SCORE* (coordinator Verstegen, Fratila-Apachitei) but will also be incorporated into joint Erasmus MC-TUD teaching programs such as Clinical Technology and Nanomedicine. Also new initiatives will be explored and aligned with the medical student curriculum (Arts 2030), and Applied Science institutions, such as Hogeschool Rotterdam and TNO Delft.

Pillar 3: creating visibility. We will present our Flagship to children and teachers by External Teaching via open workshops or seminars and Sciences School for pupils and students to engage with the next generation of scientists-to-be. The Science Gallery is an excellent opportunity to invite the general public to be engaged in biomedical sciences, art, society, education and health care. We will communicate our activities to the Public via a dedicated Twitter account, a landing page connected to the university homepages to attract young talents and involve HOUDINI students and members in output dissemination, contributions to local research days (e.g. annual Science Night), organization of sciences pitch events, and presentations at Patients Societies' meetings.

Activities to build a website and forum to share research outputs, and for education of other scientists (Houdini and beyond), clinicians, patients, students and general public, will align with existing activities within the Academic Centres "Stem Cell and Organoid Research Erasmus MC (SCORE)", Hepato-Pancreato and Biliary Diseases, Bone & Joint, AC-UT Academic centre for urogenital tumours of the EMC, the Oncotech Pillar of the Delft Health Initiative, and the Delft Bioengineering initiative. Results and findings from Houdini will be discussed with students during lectures and workgroups of the Health Technology Assessment course in the international master programs of ESHPM. Other scientists will be informed through lunch meetings and symposiums.

2.e Long term societal impact (494)

Cancer is a worldwide leading cause of death, affecting all age groups. As a rule, it is not the primary cancer that causes the ever-increasing death toll of cancer, but the spread of cancer cells to other organs that leads to 90% morbidity. Given the increasing health care and societal burdens related to metastatic disease, we need to develop therapies that focus on metastatic outgrowth. HOUDINI aims to establish a research environment at the cross-road of biomedical and technological innovation guided by societal principles that will impact metastatic cancer treatment. A holistic approach starts not with tumour cells, but with the patient. Cancer affects the somatic, psychic and

societal well-being of patients and has a broader social and economic impact. Our project is unique in that all these dimensions are addressed in unison, combining expertise from biomedical and technological fields with interactive health technology assessment and citizen ethics.
We will converge biomedical and technical sciences to improve our understanding of the role of the tumour microenvironment in the invasion of cancer cells into specific organs by combining organ-on-chip technologies with tuneable micro-environments. This will yield new human disease models that can be used for metastasis prediction studies and as platforms to screen new therapies. Moreover, the scientific value that we generate in HOUDINI will not only be relevant for metastasis of prostate and breast cancer, but also expand to other cancer types (pan-cancer perspective) burdened with bone and/or liver metastasis, such as bladder and pancreatic cancer for liver metastasis. Likewise, the organ systems of healthy human bone and liver developed in HOUDINI may well be suited to serve as starting platform to model and investigate diseased (non-cancer) liver and bone tissue. The use of such alternatives to animal models applicable in a broader biomedical research setting will make an important contribution to national and international 3R policy goals aiming to replace animal experimentation.

A preliminary literature search on the ethics of organ-on-chip technology surprisingly revealed that there is no proper ethical guidance for organ-on-chip research at the moment [1,2], especially not for cancer research. Our HOUDINI flagship has great potential in designing the blueprint of how organ-on-chip research ought to be done. A holistic approach allows to address long-term societal challenges. Patients are not seen as targets of interventions, but as active participants in the research (co-design principle), sharing their experiences, values and concerns to foster the viability, societal relevance and clinical viability of our approach. One of our long-term societal impacts will be achieved by developing guidance on the ethical aspects of research & development and implementation of potential future applications of cell-matrix technologies, including clinical utility, safety, privacy, informed consent, governance and ownership, and commercialisation. This guidance will be empirically formed by focus group studies, and aligned with the perspectives and ethical concerns of patients and the general public. By pro-actively placing this technology on the agenda of cancer research worldwide, we aim to also involve (inter)national cancer societies and EU networks.

3. Flagship proposal (25%) 1999/2000 words for 3.a to 3.d

• The Flagship program contributes to excellent Convergence Health & Technology research, at the forefront of international research (1)
• The Flagship program is both innovative as well as challenging in its research and approach (2)
• The Flagship program has a clear strategy to engage students in its research plan (3)
• The Flagship program has a clear aim and long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Coherence within the program is clearly demonstrated (4)
• The Flagship program aligns with or builds on (inter)national research agendas in this field, fit with the (inter)national research agendas but has a "high risk, high gain" main objective that is thoroughly explained (5)
• The Flagship program is feasible in terms of research performed, staff and budget needed (6)
• The Flagship program has a concrete plan for attracting external funding, which should result in at least the same amount as the central funding over the 5-year period (7)

3.a Ambitions (360 words)

HOUDINI contributes to the Convergence mission of 'life-long health for all', even when being confronted with metastatic cancer. Cancer metastasis affects the general well-being of patients hence, **HOUDINI** will start from the patient's perspective, with the ambition to establish **new prediction tools and therapies for disseminated disease, that are in line with patient's needs, and financially acceptable and feasible in clinical practice**. Current cancer therapies take a cell-centred view; however, metastasis is increasingly recognized to be dictated by a complex interplay between the normal healthy organ environment and invading tumour cells. HOUDINI aims at **targeting the whole organ metastatic niche.** HOUDINI researchers will work together on innovative technology to build human-based organ systems allowing to study the essential factors that drive metastatic invasion and growth. These developments will be complemented with societal and economical evaluations to ensure potential application and acceptance of such new technologies and knowledge by researchers, clinicians, patients and society. Participation of focus groups and potential end-users will connect natural science questions with societal needs. Together, this will further aid the long-term ambition of HOUDINI to expand its generated knowledge on two of the most common metastatic sites, bone and liver, across other research areas. Such will include the development of pan-cancer treatment applications as well as use of its overarching toolbox technology in other (non-cancer) research fields.

To build an interactive research program on holistic principles, promoting interaction and collaboration across disciplines, HOUDINI is structured in four Work Packages (WP) (Figure 2). WP 1 and 2 bring together the natural science knowledge of the biomedical and technological partners to understand and define new treatment targets for metastatic disease. WP1 is dedicated to bone (lead PI: vd Eerden) and WP2 is focussed on liver (lead PI: Verstegen). These WPs are embedded in WP3 (lead PI: Blommestein), that is dedicated to societal science providing the ethical, societal and economical aspects of the biomedical and technological innovations that are developed in WP1 and 2. Together, WP1-3 form the fundament of HOUDINI's ambition. Finally, WP4 (lead PI: vWeerden) is responsible for overall management, including coordination, education, dissemination, as well as grant writing and knowledge exchange.

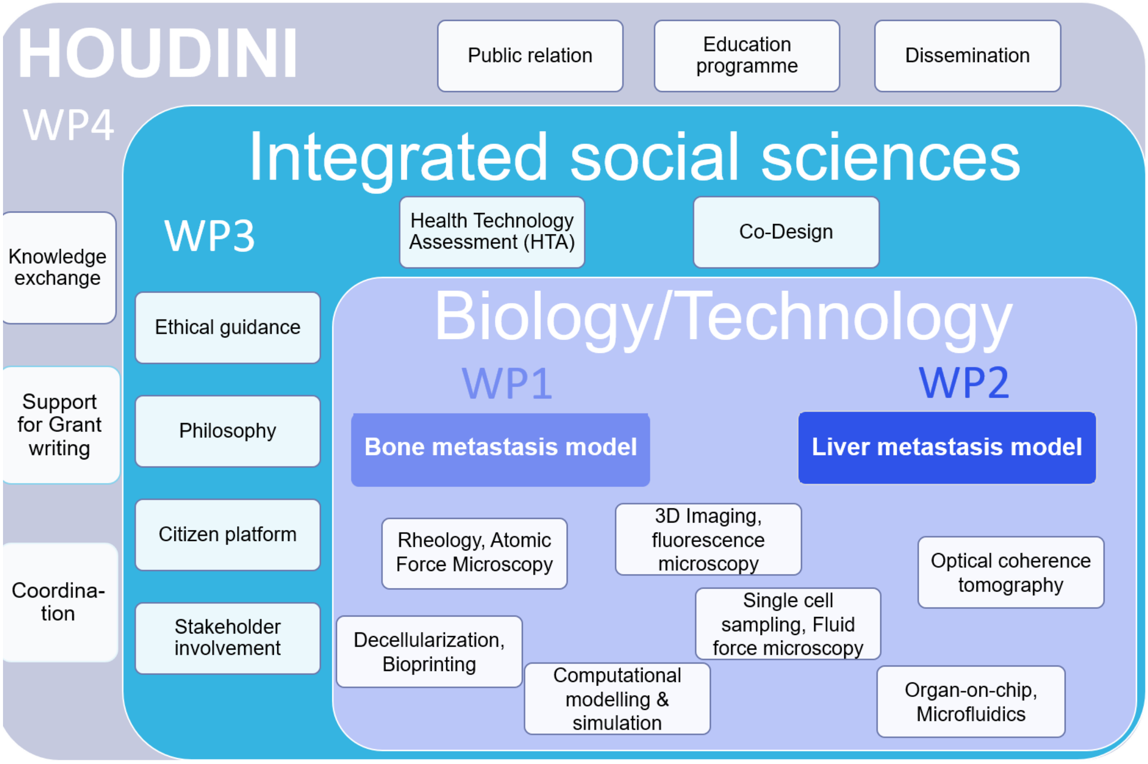

Figure 2: HOUDINI WP structure.

3.b Aims and feasibility (384 words)

HOUDINI has the ambition to develop new realistic and (economically) feasible prediction tools and treatment approaches for metastatic cancer. In HOUDINI, we uniquely combine top-down models of deconstructing/decellularizing human organs with bottom-up models, obtained by recellularizing, 3D printing and organ-on-chip approaches, to distil the crucial tissue-specific features dictating tumour cell invasion and growth. Starting with bone and liver as organs of preferential metastatic spread of prostate and breast cancer, the program will expand to a pan-cancer approach, including other cancer types and other metastatic sites.

The **HOUDINI strategy is unique** as it integrates social and natural sciences in interdisciplinary PhD projects. While the natural science PhD students will spend 3 months each on a dedicated social topic, the social PhD will learn how to translate and incorporate questions and expectations from patient's and citizens into biomedical research projects. HOUDINI will allocate a budget of 50,000 EUR per year for a Research Fund that supports pilot projects (BSc and MSc projects) to initiate and foster new creative ideas towards mature project applications. This unique interdisciplinary and innovation-driven environment will **attract and involve students** and encourage them to network and collaborate from the very beginning.

The PhD projects will be supervised by partners that have ample expertise and are embedded in a research environment with relevant facilities, guaranteeing the **feasibility of the research**. Furthermore, the HOUDINI project structure with dedicated work meetings, will assure an efficient exchange of knowledge, including sharing of dedicated protocols and facilities. On top of this, the educational program of HOUDINI will participate in existing honours programs and includes interdisciplinary workshops and seminars. To build this interactive organization, we will allocate budget for an office manager.

To **attract external funding** and build new research collaborations, the PhD and pilot projects of HOUDINI testing new research ideas and applications of developed technologies, will showcase HOUDINI to interested parties. Moreover, HOUDINI will actively stimulate PI-initiated grant applications through providing grant writing support. Together, assuming a general success rate of 25% and a matching percentage of 20% of the project budget, the **sustainability** of HOUDINI beyond year 3 will be covered. For continuous growth of HOUDINI beyond year 5, we intend to actively initiate or partner with other (inter)national parties to apply for consortium-wide research and/or network grants, such as KIC and NWA (see 3d).

3.c Work plan (895 words)

HOUDINI is dedicated to develop new realistic and (economically) feasible prediction tools and treatment options for metastatic cancer. Five initial projects (Table 1) will share knowledge, infrastructure, (core)facilities and biological/clinical materials. These PhD projects will be complemented by short-term pilot projects funded through our HOUDINI Research Fund, and focus on dedicated tasks to provide additional data.
Our data and infrastructure plan supported by local Data Stewards, includes online collaboration via Microsoft Teams, data storage at the 4TU.ResearchData and at the Research Suite/Digital Research Environment. All datasets will be accompanied by metadata to ensure access, exchange, and reproducibility.

Project 1 aims at ensuring responsible development and implementation of innovative technologies for the prediction and treatment of metastases. It embeds the biomedical/technological **projects 2-5** in an ethical and societal context. From the start, a constructive public dialogue with cancer patients and the public will be organized about the opportunities (e.g., animal model alternatives, co-design) and challenges (e.g., tissue donation) of such technologies. Additionally, early Health Technology Assessment of costs associated with developments of projects 2-5 will be performed.
Projects 2-5 investigate the tissue microenvironments' role in cancer cell invasion into bone and liver as rigid and soft tissues. Prostate and breast cancer frequently spread to these tissues and therefore, all projects start with established prostate and breast cancer cell lines as surrogates of circulating tumour cells (CTCs), followed by validation with CTCs from biobanks, and fresh CTCs provided by our clinical partners.
Projects 2 and 3 will develop bone models. **Project 2** focuses on 3D-bioprinted human bone scaffolds to define how ECM composition and biomechanics (AFM, nanoindentation, rheology) affect tumour cell adhesion and migration (confocal microscopy; link projects 4 and 5). In parallel, **Project 3** investigates tumour cell extravasation using organ-on-chip (OoC) technology (link Project 5). The cancer cell-induced biomechanical changes of bone ECM will be determined through rheology and AFM (link Project 4) and invasion of cancer cells will be monitored by time-lapse imaging, computational modelling and optical coherence tomography. Immune cells, such as macrophages (link Project 5), and extracellular vesicles (EVs), will be introduced investigating their impact on tumour cell behaviour.
Projects 4 and 5 model liver metastases. **Project 4** focuses on the role stromal cells in supporting tumour cell invasion. Human decellularized livers will be biomechanically characterized (AFM, nanoindentation, rheology) and re-cellularized with stromal fibroblasts and cancer cells. The fibroblast-complemented ECM changes and interactions with tumour cells will be profiled by SILAC spectrometry, scRNA-sequencing and 4D-confocal microscopy. Macrophages (link Project 5) will be included to increase model complexity (link Project 2). **Project 5** analyses the macrophage response to tumour cell adhesion and invasion. We will modify a liver OoC containing a hepatocyte and endothelial layer (link Project 3) with macrophages. The biomechanical forces of the blood flow on macrophages and adhesion of macrophages and cancer cells to the endothelium will be measured

using advanced fluorescent microscopy (link Project 3), fluid force microscopy and AFM (link Project 4).

In the first phase (years 1-5), HOUDINI will establish human bone and liver metastasis toolbox models supported by citizens and patients' input. New knowledge on the determinants of metastasis with special focus on how ECM, endothelial, stromal, and immune cells affect tumour cell behaviour will be collided to initiate new predictive and therapeutic approaches of metastatic disease thus representing a first milestone of HOUDINI's long-term mission.

Table 1: HOUDINI projects. (Fluid force microscopy, FFM; Atomic force microscopy, AFM; Optical Coherence Tomography, OCT; Health Technology Assessment, HTA)

Project	**1: EUR**	**2: EMC**	**3: TUD**	**4: TUD**	**5: EMC**
	Societal & economical aspects of holistic approaches for metastasis treatment	**Tumour-bone cell interactions in 3D bioprinted bone-ECM scaffold**	**Tumour cell behaviour under shear stress in bone-on-chip**	**Tumour-stromal complexity of liver**	**Mechanics of immune cell-assisted tumour cell invasion of liver**
Starting project leads	Blommestein, Bunnik, Zwart, Leeneman	vd Eerden/ Fratila-Apachitei/ Zadpoor/ vRoyen	Boukany/ vd Eerden/ Kenjeres/ Kalkman	Verstegen/ Koenderink/ vRoyen	vWeerden/ Linke/ Dik/ Leenen/ Ghatkesar
Focus point	Ethics & Health Technology Assessment	Extracellular matrix	Endothelial cells	Stroma cells (fibroblasts)	Immune cells (macrophages)
Scientific topic year 1- 2	Exploring available and acceptable treatment options for patients	Characteristics of bone metastasis ECM	Endothelial barrier crossing of cancer cells in flow	Stromal cell features in decellularized human liver	Macrophage effect on cancer cell's adhesion in flow
Toolbox platform year 1-2	Citizen platform, ethical analysis, early HTA	Bioink development and bone scaffold bioprinting	Bone-on-chip and OCT- flow characterisation	Decellularized liver and stromal cell profile (AFM, rheology, proteomics)	Macrophage-cancer cell adhesion (FFM, AFM)
Holistic knowledge year 1-2	Responsible development and implementation of holistic approaches	Biomechanical ECM features for cancer cell interaction	Biomechanics and endothelial cell dynamics	Stromal cell proteome landscape in decellularized liver	Macrophage polarisation impacting cancer cell landing
Scientific topic year 3-4	Patients' values, expectations and concerns regarding novel treatments	Bone ECM remodelling by bone and cancer cells, and their EVs	Interaction dynamics between cancer cells and other bone cell types	Cancer cell invasion into decellularized liver co-cultured with stromal cells	Macrophage effect on cancer cell invasion into the Space of Disse
Toolbox platform year 3-4	Joint PhD research activity, educational sessions	AFM and 3D-tumour cell imaging in scaffold	Rheology, imaging and computational modelling	Single cell imaging and retrieval	Single cell retrieval (FFM) and migration under flow
Holistic knowledge year 3-4	Use of technologies in line with ethical values and societal preferences	Metastatic cancer cell behaviour in remodelled bone ECM	Dynamics of cancer cell invasion under flow in bone-on-chip	Characteristics of premetastatic liver niche	Macrophage characteristics supporting cancer cell liver invasion

3.d Alignment of the program with relevant national and international research agendas (360 words)

The HOUDINI program aligns with the concepts of "*Responsible Research and Innovation*" and "*open science*" that aim to make research more inclusive and responsive by involving societal and patient values and perspectives. Both notions are guiding concepts for research programs such as **Horizon Europe** and the **National Science Agenda (NWA)**.

NWA made a strong plea for defining the right treatment with the fewest possible side effects for each patient. With cancer being the leading cause of death in the Netherlands, HOUDINI aligns particularly to this program. Other NWA defined routes, such as 'Health care research, sickness prevention and treatment,' 'Measuring and detecting' and 'Regenerative medicine' are also well aligned with HOUDINI. This latter theme as well as the national **'Transitie Proefdiervrije Innovatie (TPI)'** or the European-wide initiative on animal-free innovations (**European Partnership for Alternative Approaches to Animal Testing**), aim to reduce lab animal use, that fits the HOUDINI toolbox development. HOUDINI's structure is well-suited to converge multidisciplinary stakeholders that help to increase (public) acceptance and application of such innovative animal-free systems. Already some HOUDINI partners are involved in the NWO 'Human measurement model' research program.
Additionally, HOUDINI has multiple opportunities to participate in the long-term ambition of the **Dutch Cancer Society KWF** as defined in 'Ambitie 2030' to improve better quality of life for patients with and without cancer, as well as in joined **NWO Perspective** calls on 'New technology through consortia'.

Within the new Horizon Europe program, the EU launched the **European Mission on Cancer** together with the '**Europe's Beating Cancer Plan**.' The HOUDINI consortium is an attractive partner to join European consortia and an excellent starting point to initiate training of a new generation of expert researchers in the context of the **Marie Curie Innovative Training Networks**. HOUDINI is dedicated to set-up and share digital infrastructures in support of research and to bring value to the new **Knowledge Centre on Cancer** and the **European Cancer Information System**, and to contribute to Horizon Europe partnerships and **European Institute Innovation & Technology** and **Knowledge and Innovation Communities** in the health and digital areas. This will substantiate HOUDINI outcome towards impactful actions and accelerate translation into public health and clinical practice.

4. Consortium (25%) 1785/2000 words for 4.a to 4.d

• The Flagship program is driven and executed by a well-organized team of scientists from all three institutions and at least 4 departments or faculties in total. In exceptional cases, it is possible to form a Flagship program with only two leads from two institutions, but this should be clearly motivated (1)
• Members of the consortium have a track record of excellence in academic research, translation of scientific results into clinical application or societal impact (2)
• Members of the consortium have a significant track record in engaging and supervising BSc and MSc students in research projects (3)
• Members of the consortium have demonstrated to be able to attract external funding, thereby enhancing the programs sustainability after the 5 years of central funding (4)
• There is a demonstrable fit between the aims and objectives of the program and the expertise of the consortium members (5)
• End users (private and/or public sectors) are involved from an early start and team members are well-connected to external partners in the ecosystem (6)
• Governance should be clearly described and is organized in such a way that it facilitates collaboration and optimally makes use of the expertise of all participants (7)

4.a Track record of the consortium members (564 words)

All members of HOUDINI have published together more than 2,000 papers, and supervised in the last 5 years 112 PhD and 276 Master students. Our members have a strong track record in translational research as well as for integrating societal and ethical issues in early-stage research and technology development (RRI, ELSA, citizen ethics). In addition to the publication record this is also displayed in current grant money that is related to HOUDINI comprising more than 7.8 million EUR funding in total (see individual information in Table 2). From this, we are confident that matching in the coming years is realistic.

Table 1: Overview of track record of HOUDINI members

MEMBER	RECORD WEB OF SCIENCE 02/22: PUBLICATIONS & H INDEX	CURRENT GRANTS RELATED TO HOUDINI	STUDENTS IN PAST 5 YEARS (PHD/MSC)
WYTSKE VAN WEERDEN	183, H=39	Health~Holland/SGF (2020-2023) 635,000 EUR, Building a multi-tissue microfluidics system of metastatic potential (BIOMEP): 3D PCa metastatis models	7/7
GIJSJE KOENDERINK	167, H=44	ENW-XL (The Active Matter Physics of Collective Metastasis): 371,000 EUR: *in vitro* models for metastasis and biomechanical analysis	8/24
HUB ZWART	98, H=12	EU Horizon 2020, joinus4health (cohort studies and citizens science): 195,000 EUR, International Human Microbiome CSA (Ethics WP): 88,000 EUR, NWO, BaSyC (building a synthetic cell), 2 PhD projects	10/5
MONIQUE VERSTEGEN	64, H=24	HDMA (Erasmus MC) 2018; 230,000 EUR: Assessing tumour microenvironment in bile duct cancer. / ENW-XS 2021; 50,000 EUR: bile duct cancer on a chip/Convergence Open Mind	7/20

		2021; 10,000 EUR -expertise: ECM biomechanics in bile duct tumour metastases	
BRAM VAN DER EERDEN	102, H=27	Health~Holland-TKI (PhosphoNorm): 929,000 EUR, use of human bone tissue for bone cell analyses.	12/5
MARTIJN LOLKEMA	206, H=40	NWO/KWF (2020-2024)300,000 EUR, PICTURES: Probing Intercellular heterogeneity in Circulating TUmor cells of de novo metastatic Hormone Sensitive Prostate CanceR patiEntS, Co-Project leader	5/0
WIM DIK	142, H=29	-	11/4
PIETER LEENEN	151, H=55	-	3/10
ELINE BUNNIK	58, H=14	KWF (2020-2024) 500,000 EUR, Ethics of access to cancer treatments, EU Horizon 2020 (2020-2024) 250,000 EUR Ethics of early-phase clinical research in regenerative medicine (VANGUARD project), ZonMw (2022-2026) 250,000 EUR, Ethics of development of patient-derived cell models for personalised medicine	5/15
ARNO VAN LEENDERS	19, H=4	-	10/1
MARTIN VAN ROYEN	63, H=23	Health~Holland/SGF BIOMEP (2020-2023) 635,000 EUR, (PI Wytske van Weerden); Marie Curie ITN ProEVLifeCycle (PI: GUIDO JENSTER) (2020-2024) 250,000 EUR, KWF Alpe d'HuZes Unieke Kansen IMMPROVE (2016-2022) 279,000 EUR: Analysis of Extracellular Vesicles; Prostate Cancer UK (2013-2015) 117,000 EUR, Extracellular Vesicle analysis and biology	3/7
POUYAN BOUKANY	56, H=27	ERC-COG (2019-2024) 2,000,000 EUR: FANCY: Flow and Deformation of Cancer tumours near Yielding	5/25
JEROEN KALKMAN	66, H=25	TTW-Perspectief 2017 Synoptics P6 250,000 EUR, Flow imaging; HTSM 316,000 EUR, 2019 NanorheoOCT	4/4
MURALI GHATKESAR	47, H=13	Convergence grant (2020-2022) 150,000 EUR, Single cell biopsy	0/15
AMIR ZADPOOR	273, H=55	Interreg 2Seas- SiteDrug (2020-2023) 479,000 EUR	10/50
LIDY FRATILA-APACHITEI	72, H=25	RMS foundation- Role of 3D scaffold architecture (2021-2023) 140,000	3/15
SASA KENJERES	148, H=33	Dutch Heart Foundation, (2018-2022) 20,000 EUR,"RADAR"; FF4EuroHPC, (2022-2023) 25,000 EUR: mathematical modelling and computer simulations of biological/biomedical transport phenomena	8/62

HEDWIG BLOMMESTEIN	66, H=10	-	5/17
FRANZISKA LINKE	16, H=3	-	1/5
PEI-HUA HUANG	5, H=3	Open Mind Call for Health and Technology, 2021 10,000 EUR	-
BRENDA LEENEMAN	31, H=6	-	0/8
PIETER VAN DE KERCKHOVE	3, H=2	Educational Grant from Erasmus University (2020-2021) 20,000 EUR: interdisciplinary course on design considering societal aspects	0/5

4.b Expertise of the consortium members in relation to the aims and objectives of the program (601 words)

To achieve our set aims and milestones, HOUDINI will build on existing research being conducted by each research partner individually or within collaborations of the three institutions. Establishing larger alliances has been hampered by a lack of (formal) connections and meetings to inform and educate each other on the existence, innovations and/or potentials of the ongoing activities in each institution. The HOUDINI consortium embraces an effective and inspiring environment of interdisciplinary research and education to target disseminated cancer. Each consortium partner has been selected based on their unique expertise that fits the objectives of the HOUDINI project. In Table 3 the complementary research expertise and contribution of each individual partner is indicated. The consortium brings together experts from cancer biology, liver and bone tissue engineering, biomechanics, imaging, microfluidics, computational modelling, immunology, pathology, health technology, ethics, philosophy and the clinic.

Table 2: Overview of HOUDINI consortium members, their research expertise and contribution to HOUDINI

	RESEARCH EXPERTISE [HOUDINI--RELEVANT PUBLICATIONS]	CONTRIBUTION
WYTSKE VAN WEERDEN (CANCER BIOLOGIST)	(prostate) cancer metastasis modelling, cancer mechanics/proteomics/signalling [3–5]	Cancer-induced matrix remodelling, cancer cell-endothelial/macrophage interactions
GIJSJE KOENDERINK (BIOPHYSICIST)	(cancer) cell mechanics, tissue reconstitution, cytoskeleton, matrix [6–8]	Matrix and cell biomechanics, cell migration, biophysical tools, microfluidics, AFM
HUB ZWART (PHILOSOPHY)	philosophical and ethical issues in life sciences [9–11]	Focus group/patient perspective and involvement
MONIQUE VERSTEGEN (BIOLOGIST)	liver transplantation, tissue engineering liver primary cancer, matrix [12]	Human liver acquisition, decellularized liver ECM, Liver-on-Chip, drug screening
BRAM VAN DER EERDEN (CELL BIOLOGIST)	extracellular bone matrix, bone tissue engineering, organ-on-chip [13,14]	Human bone acquisition, bone ECM, drug screening
MARTIJN LOLKEMA (MED. ONCOLOGIST)	Large scale genomic analyses of metastatic cancer tissues, translational biomarker studies, early phase drug development [15–17]	Validation in clinical samples, sample acquirement

WIM DIK (MEDICAL IMMUNOLOGIST)	innate and adaptive immunology, inflammation, fibroblast role in inflammation, [18–20]	Immunology, immunotyping, endothelial-immune cell interactions
PIETER LEENEN (IMMUNOLOGIST)	innate macrophage biology, inflammation, interaction with bone cartilage [21–23]	Immunology, immunotyping, endothelial-macrophage interactions
ELINE BUNNIK (MEDICAL ETHICS)	Medical ethics, bioethics [24–26]	Ethical assessment of technology, governance of data, access, patient perspective
ARNO VAN LEENDERS (PATHOLOGIST)	(prostate) cancer, tissue growth patterns and 3D structure [27–29]	Human FFPE samples and biobank, pathological expertise on morphology of metastatic lesions
MARTIN VAN ROYEN (MOLECULAR BIOLOGIST)	Extracellular vesicle (EV) biology, EV analysis, 3D-tissue imaging [5,30,31]	EV biology, quantitative confocal microscopy, tissue clearing,
POUYAN BOUKANY (MICROFLUIDICS)	Biomicrofluidics and soft matter mimicking of tissues, organ-on-chip, 3D cell migration/cell invasion with relevant TME [6,32]	TME on-a-chip, organ-on-chip models, biophysical characterization
JEROEN KALKMAN (PHYSICIST)	optical tomography tools and applications [33,34]	Optical coherence tomography, flow and tissue structure/mechanical read-outs
MURALI GHATKESAR (MICRO AND NANO ENGINEERING)	micro/nanomechanical microfluidics tools, biophysical characterisation [35–37]	Cell-cell/matrix adhesion, matrix stiffness, single cell analysis
AMIR ZADPOOR (BIOPRINTING)	manufacturing biomaterials at nanoscale, bioprinting, organ-on-chip [38,39]	3D/4D (bio)printing, biomechanical characterization, physical cues, computational modelling
LIDY FRATILA-APACHITEI (BIOMATERIALS)	biomedical materials, instructive materials [40,41]	3D/4D (bio)printing, biomechanical characterization, physical cues, computational modelling
SASA KENJERES (COMPUTER MODELLING)	microfluidics, fluid dynamics, fluid mechanics, computer modelling and simulation [42–44]	Cancer cell flow modelling and simulation
HEDWIG BLOMMESTEIN (HTA)	Health Technology Assessment (HTA) [45–47]	Health Technology Assessment (HTA), early conceptual HTA modelling
FRANZISKA LINKE (CANCER BIOLOGIST)	cancer metastasis modelling, cancer signalling, cancer matrix and macrophage interaction [48–50]	Cancer-induced matrix remodelling, cancer cell-endothelial/macrophage interactions
PEI-HUA HUANG (ETHICS)	Ethics [51,52]	Ethical assessment, governance of data, availability of technology, patient perspective

BRENDA LEENEMAN (HTA)	Health Technology Assessment; Clinical epidemiology; Costing studies; Oncology (advanced disease) [53–55]	Health Technology Assessment (HTA)
PIETER VAN DE KERCKHOVE (CO-DESIGN)	Collaborative design and design thinking methodology to develop digital health products and services [56–58]	Co-design, patient involvement

4.c Involvement of partners (270 words)

The HOUDINI consortium is already well-connected with patient organizations, industry partners and non-profit organizations. A unique aspect of our approach is the early involvement of patients and citizens in the fundamental science devoted to the development of novel metastasis prediction tools and therapy approaches. The national liver and prostate cancer patient organizations (Nederlandse Leverpatienten Vereniging, NLV; ProstaatkankerStichting) strongly support our Flagship. The patients will be a stakeholder in ethical evaluations and informing the (patient) society with novel insights into possible future therapeutic options. EUR members (Zwart) are also closely connected to the Rathenau Institute, which can help us organize stakeholder sessions to discuss patient-centred holistic biomedicine.

Several partners from EMC and TUD (vWeerden, Verstegen, vRoyen, Kalkman) are already working with microchips and microfluidic setups from Bi/OND (Delft, The Netherlands). This partnership will be deepened during this Flagship, contributing to both liver and bone research within the different HOUDINI projects (see attached support letter). Another running project (vWeerden) is collaborating with InSphero (Schlieren, Switzerland) and will include their primary liver microtissues in project 5. Additionally, partners from EMC (vWeerden) have a long-standing collaboration with leading pharmaceutic companies on preclinical compound testing. These companies have endorsed the European-wide initiative on animal-free innovations "European Partnership for Alternative Approaches to Animal Testing (EPAA) and are likely interested in incorporating our new technological developments in future collaborative projects.

Multiple HOUDINI members are also members of the Health~Holland "human Organ and Disease Model Technologies, hDMT" consortium, which serves as a forum for Organ-on-Chip collaborations between public and private partners. hDMT will support HOUDINI by co-organization of meetings, workshops and dissemination of results from the very beginning.

4.d Governance, organization and collaboration strategy of the consortium (350 words)

The organization and management tasks of HOUDINI are included in **WP4**. The management of HOUDINI is governed by a **steering committee** that is formed by all research consortium partners and that will guide the (operational) progress of the project from a scientific perspective. The 3 lead PIs of the 3 participating institutions will jointly act as chairs of HOUDINI. A **scientific advisory board** will be installed that includes not only all research consortium partners, but also experts of the three disciplines from outside the three institutions, to discuss scientific progress and may advise on (new) research directions.
A **user committee** will be consulted regularly consisting of relevant stake-holders and/or end-users. This committee will focus on the progress/choices in the project in relation to the expectations of

the users in order to maximize applicability of the project outcome. The organisation of committee meetings will be the task of a **HOUDINI office** employed with a program support officer (0.4 FTE). This position may be expanded to 0.6-0.8 FTE during expansion of HOUDINI's activities. The HOUDINI office will support collaboration, education and communication through the following planned activities. In year 1, WP4 will organize a **live kick-off meeting** of the entire consortium. From year 2 onwards, an open **annual HOUDINI symposium** will be organized with HOUDINI PIs, PhDs and master students presenting their work, to reach-out and attract new research parties, as well as inviting external researchers to ensure communication with other researchers and institutes.
Semi-annual meetings are planned for HOUDINI participants to share data and discuss new plans and funding initiatives. Additionally, **quarterly focus group meetings** will be organized for dedicated discussions on specific topics to assure optimal and timely exchange of results and new findings across WPs within the consortium.
To provide training and education on various aspects of HOUDINI's portfolio, we will setup **dedicated workshops** and a **seminar series** in connection with local cancer-related initiatives, e.g Oncotech (part of the Delft Health Initiative of TU Delft) and the Erasmus MC Cancer Institute.
These activities will be communicated through a quarterly **newsletter,** advertising upcoming meetings, new employees, grant opportunities and successes.

Annexes

I Budget table (including the external funding table)

II Letters of intent and letters of commitment

From BI/OND Solutions EV, InSphero AG, human Organ and Disease Model Technologies (hDMT), ProstaatkankerStichting, Nederlandse Leverpatienten Vereniging

III Letter of support from department head of the main lead

IV References

Budget formulier: Convergentie Flagship Call

Flagship:	Holistic approach to understand and target the metastatic niche
Flagship acronym:	HOUDINI
Naam Penvoerder:	Wytske van Weerden
Startdatum en looptijd van het project:	01/11/2022 - 31/10/2027
Totaal project budget:	4,000,000 EUR

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.

Loonkosten:								2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	**Functie**	**Aantal fte**	**Startdatum**	**Einddatum**	**Inschaling**	**Maandbedrag**	**Kosten**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	
Onderzoeksorganisaties																						
EMC Wytske van Weerden Matching	Postdoc 1	0.4	01/11/2022	31/10/2024	Postdoc 1	€ 6,998	€ 67,179	€ 5,441		€ 32,960		€ 28,778								€ 67,179	€ 0	€ 67,179.00
EMC Wytske van Weerden Matching	PhD 1	0.2	01/11/2022	30/10/2026	PhD 1	€ 3,613	€ 52,030	€ 1,951.50		€ 11,850		€ 12,694		€ 13,555		€ 11,980.83				€ 52,031	€ 0	€ 52,031.00
EMC Monique Verstegen Matching	PhD 2	0.4	01/11/2022	30/06/2023	PhD 2	€ 3,219	€ 15,452	€ 3,863		€ 11,589										€ 15,452	€ 0	€ 15,452.00
EMC Monique Verstegen Matching	Reserach Tech 1	0.4	01/11/2022	30/10/2026	Research tech 1	4173.65	€ 117,698	€ 4,720		€ 28,532		€ 29,772		€ 30,422		€ 24,251.67				€ 117,698	€ 0	€ 117,697.67
EMC Bram van der Eerden Matching	Reserach Tech 2	0.3	01/11/2022	30/06/2023	Research tech 2	€ 4,671	€ 14,014	€ 3,504		€ 10,511										€ 14,014	€ 0	€ 14,014.00
EMC Bram van der Eerden Matching	PhD 3	0.2	01/11/2022	30/06/2023	PhD 3	€ 3,219	€ 7,726	€ 1,932		€ 5,795										€ 7,726	€ 0	€ 7,726.00
EMC Martin van Royen Matching	PhD 4	0.3	01/11/2022	31/10/2024	PhD 4	€ 3,370	€ 36,394	€ 2,927		€ 17,775		€ 15,692								€ 36,394	€ 0	€ 36,394.00
EMC Eline Bunnik Matching	PhD 5	0.6	01/11/2022	31/10/2024	PhD 5	€ 3,370	€ 72,788	€ 5,855		€ 35,549		€ 31,384								€ 72,788	€ 0	€ 72,788.00
EMC Eline Bunnik Matching	Postdoc 2	0.2	01/11/2022	31/10/2024	Postdoc 2	€ 4,548	€ 32,748	€ 2,648		€ 16,051		€ 14,049								€ 32,748	€ 0	€ 32,748.00
TUD Gijsje Koenderink Matching	PhD	0.2	01/11/2022	31/10/2024	PhD	€ 3,102	€ 22,334	€ 1,861		€ 11,167		€ 9,306								€ 22,334	€ 0	€ 22,333.54
TUD Pouyan Boukany Matching	PhD	0.2	01/11/2022	31/10/2024	PhD	€ 3,102	€ 22,334	€ 1,861		€ 11,167		€ 9,306								€ 22,334	€ 0	€ 22,333.54
TUD Jeroen Kalkman Matching	PhD	0.4	01/11/2022	31/10/2024	PhD	€ 3,102	€ 44,667	€ 3,722		€ 22,334		€ 18,611								€ 44,667	€ 0	€ 44,667.07
TUD Amir Zadpoor Matching	PhD	0.2	01/11/2022	31/10/2024	Postdoc	€ 4,209	€ 30,304	€ 2,525		€ 15,152		€ 12,627								€ 30,304	€ 0	€ 30,303.86
TUD Amir Zadpoor Matching	postdoc	0.2	01/11/2022	31/10/2024	PhD	€ 3,102	€ 22,334	€ 1,861		€ 11,167		€ 9,306								€ 22,334	€ 0	€ 22,333.54
TUD Lidy Fratila-Apachitei Matching	PhD	0.2	01/11/2022	31/10/2024	PhD	€ 3,102	€ 22,334	€ 1,861		€ 11,167		€ 9,306								€ 22,334	€ 0	€ 22,333.54
TUD Sasa Kenjeres Matching	PhD	0.1	01/11/2022	31/10/2024	PhD	€ 3,102	€ 11,167	€ 931		€ 5,583		€ 4,653								€ 11,167	€ 0	€ 11,166.77
EUR Hub Zwart Matching	PhD	1.0	01/11/2023	31/01/2024	PhD-EUR	€ 3,189	€ 14,351			€ 9,567		€ 4,784								€ 14,351	€ 0	€ 14,351.06
EUR Hedwig Blommestein Matching	PhD	0.2	01/11/2022	31/10/2024	PhD-EUR	€ 3,189	€ 22,961	€ 1,913		€ 11,481		€ 9,567								€ 22,962	€ 0	€ 22,961.70
HOUDINI anticipated grants Matching	PhD	2.8	01/01/2025	31/10/2027	PhD-Cons	€ 3,613	€ 515,966							€ 182,106		€ 182,106		€ 151,755		€ 515,966	€ 0	€ 515,966.39
HOUDINI anticipated grants Matching	Postdoc	0.4	01/11/2024	31/10/2027	Postdoc Cons	€ 6,998	€ 151,152					€ 8,397		€ 50,384		€ 50,384		€ 41,987		€ 151,152	€ 0	€ 151,152.48
HOUDINI anticipated grants Matching	Research Tech	0.4	01/11/2024	31/10/2027	Research Tec Cons	€ 4,671	€ 100,899					€ 5,606		€ 33,633		€ 33,633		€ 28,028		€ 100,899	€ 0	€ 100,899.00
HOUDINI consortium	theme support office	0.4	01/11/2022	31/10/2027	Theme supp off	€ 3,741	€ 134,672	€ 3,903		€ 23,700		€ 25,386		€ 27,110		€ 29,063		€ 25,511		€ 134,672	€ 0	€ 134,672.00
HOUDINI consortium	Postdoc	2.0	01/01/2027	31/10/2027	Postdoc Cons	€ 6,998	€ 209,940											€ 209,940		€ 209,940	€ 0	€ 209,940.00
HOUDINI consortium	PhD	5.0	01/11/2022	31/10/2026	PhD	€ 3,613	€ 1,300,756	€ 48,789		€ 296,249		€ 317,325		€ 338,879		€ 299,516				€ 1,300,757	€ 0	€ 1,300,757.00
Ondernemingen																						
Onderneming 1						€ -	€ 0													€ 0	€ 0	€ 0.00
Overige (private) partners																						
Naam 1						€ -	€ 0													€ 0	€ 0	€ 0.00
						Totaal	**€ 3,042,201**														**Totaal loonkosten**	**€ 3,042,201**

2.

Kosten van materialen en hulpmiddelen:				2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	**Materiaal / hulpmiddel**		**Kosten**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	
Onderzoeksorganisaties																		
EMC Matching	Transitievergoeding		€ 15,000											€ 15,000.00		€ 15,000	€ 0	€ 15,000.00
EMC Wytske van Weerden Matching	BIOMEP liver spheroids		€ 10,000	€ 833.33		€ 5,000		€ 4,166.67								€ 10,000	€ 0	€ 10,000.00
EMC Monique Verstegen Matching	15K euro (ENW-XS)		€ 15,000	€ 1,250.00		€ 7,500		€ 6,250.00								€ 15,000	€ 0	€ 15,000.00
EMC Martin van Royen Matching	10k (EV imaging tools and analysis (EVQuant)		€ 10,000		€ 833.33		€ 5,000		€ 4,166.67							€ 0	€ 10,000	€ 10,000.00
TUD Jeroen Kalkman Matching	TTW-Perspectief 2017 Synoptics P6: materials matching: 10.000 Euro & HTSM 2019 NanorheoOCT mater		€ 20,000		€1,666.67		€10,000.00		€ 8,333.33							€ 0	€ 20,000	€ 20,000.00
TUD Sasa Kenjeres Matching	CPU/GPU computational cluster equivalent to 100kEuro per year		€ 117,724		€ 2,541.42		€31,849.00		€ 83,333.33							€ 0	€ 117,724	€ 117,723.75
TUD Amir Zadpoor Matching	Amir Zadpour SiteDrug grant		€ 3,500		€ 333.33		€1,916.67		€ 1,250.00							€ 0	€ 3,500	€ 3,500.00
TUD Lidy Fratila-Apachitei Matching	RMS foundation Role of 3D scaffold architecture grant		€ 10,476		€ 698.33		€4,539.33		€ 5,238.33							€ 0	€ 10,476	€ 10,476.00
HOUDINI anticipated grants Matching	Material for biotechnological lab work		€ 186,536						€ 3,792.00		60457.00		€ 62,287.00		€ 60,000.00	€ 0	€ 186,536	€ 186,536.00
HOUDINI consortium	Equipment for model development and biomechanical measurements		€ 30,000			€ 20,000								€ 10,000.00		€ 30,000	€ 0	€ 30,000.00
HOUDINI consortium	Material for dissemination, education, outreach activities		€ 25,052			€5,052.00		€5,000.00				€5,000.00		€10,000.00		€ 25,052	€ 0	€ 25,052.00
HOUDINI consortium	Master seed money (10x5k per year)		€ 242,345	€ 8,333.33		€ 50,000		€ 50,000		€ 34,011		€ 50,000		€ 50,000.00		€ 242,345	€ 0	€ 242,344.66
Ondernemingen																		
BIOND Matching			€ 9,000		€ 300.00		€ 1,800		€ 1,800.00		€ 1,800.00		€ 1,800.00		€ 1,500.00	€ 0	€ 9,000	€ 9,000.00
Overige (private) partners																		
Naam 1																€ 0	€ 0	€ 0.00
		Totaal	**€ 694,632**														**Totaal kosten materialen en hulpmiddelen**	**€ 694,632**

3.

Kosten van gebruik van machines en apparatuur							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	**Machine / apparatuur**			**Prijs per gebruikseenheid**	**Gebruiks eenheden**	**Kosten**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	
Onderzoeksorganisaties																					
EMC Wytske van Weerden Matching	Akura Flow chip system			€ 5,000	€ 4	€ 20,000	€ 1,666.67		€ 10,000		€ 8,333.33								€ 20,000	€ 0	€ 20,000.00
EMC Monique Verstegen Matching	perfusion equipment for large organs / microfluidic Organoplate Gra			€ 10,000	€ 5	€ 50,000		€ 4,166.67		€ 25,000.00		€ 20,833.33							€ 0	€ 50,000	€ 50,000.00
HOUDINI anticipated grants Matching	Chips and perfusion systems					€ 34,599								5643		€ 11,558.00		€ 17,398.00	€ 0	€ 34,599	€ 34,599.00
Ondernemingen																					
Onderneming 1						€ 0													€ 0	€ 0	€ 0.00
Overige (private) partners																					
Naam 1																			€ 0	€ 0	€ 0.00
					Totaal	**€ 104,599**														**Totaal kosten machines en apparatuur**	**€ 104,599**

4.

Aan derden verschuldigde kosten				2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	**Naam derde, omschrijving kosten**		**Kosten**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	
Onderzoeksorganisaties																		
HOUDINI consortium	Meeting/Workshop/conference/homepage organization incl catering		€ 57,234	€ 5,641.83		€ 5,000		€ 2,288.83				€ 16,421.17		€ 27,882.00		€ 57,234	€ 0	€ 57,233.83
Ondernemingen			€ 0															
Onderneming 1																€ 0	€ 0	€ 0.00
Overige (private) partners																		
hDMT Matching	Co-organization of meetings, workshops and Dissemination of results		€ 10,000		€ 333.33		€ 2,000.00		€ 2,000.00		€ 2,000.00		€ 2,000.00		€ 1,666.67	€ 0	€ 10,000	€ 10,000.00
		Totaal	**€ 67,234**														**Totaal kosten derden**	**€ 67,234**

5.

Publicatie, reis- en verblijfkosten				2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	**Naam derde, omschrijving kosten**		**Kosten**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	
Onderzoeksorganisaties																		
EMC Monique Verstegen Matching	travel: 1 int conference for 1 PhD student (~3K euro) per year (so 4x3K-12K euro)		€ 12,000		€ 1,000.00		6000		€ 5,000.00							€ 0	€ 12,000	€ 12,000.00
EMC Bram van der Eerden Matching	Fund money available for publication fees and conference visit (5K per year)		€ 10,000		€ 833.33		5000		€ 4,166.67							€ 0	€ 10,000	€ 10,000.00
TUD Gijsje Koenderink Matching	Fund money available for publication fees and conference visit (5K per year)		€ 10,000		€ 833.33		5000		€ 4,166.67							€ 0	€ 10,000	€ 10,000.00
HOUDINI anticipated grants Matching	Fund money available for publication fees and conference visit (20K per year)		€ 59,333						€ 3,333.33		€ 20,000.00		€ 20,000.00		€ 16,000.00	€ 0	€ 59,333	€ 59,333.33
Ondernemingen																		
Onderneming 1																€ 0	€ 0	€ 0.00
Overige (private) partners																		
Naam 1																€ 0	€ 0	€ 0.00
		Totaal	**€ 91,333**														**Totaal kosten derden**	**€ 91,333**

6.

Totale kosten	**€ 4,000,000**	**€ 4,000,000**

FINANCIERING - Bijdragen per partner per jaar

	in Euro	2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
1	EMC Wytske van Weerden Matching	€ 9,892		€ 59,810		€ 53,972		€ 13,555		€ 11,981				€ 149,210	€ 0	€ 149,210
2	EMC Matching											€ 15,000		€ 15,000	€ 0	€ 15,000
3	EMC Monique Verstegen Matching	€ 9,833	€ 5,167	€ 47,621	€ 31,000	€ 36,022	€ 25,833	€ 30,422		€ 24,252				€ 148,150	€ 62,000	€ 210,150
4	EMC Bram van der Eerden Matching	€ 5,435	€ 833	€ 16,305	€ 5,000		€ 4,167							€ 21,740	€ 10,000	€ 31,740
5	EMC Martin van Royen Matching	€ 2,927	€ 833	€ 17,775	€ 5,000	€ 15,692	€ 4,167							€ 36,394	€ 10,000	€ 46,394
6	EMC Eline Bunnik Matching	€ 8,503		€ 51,600		€ 45,433								€ 105,536	€ 0	€ 105,536
7	TUD Gijsje Koenderink Matching	€ 1,861	€ 833	€ 11,167	€ 5,000	€ 9,306	€ 4,167							€ 22,334	€ 10,000	€ 32,334
8	TUD Pouyan Boukany Matching	€ 1,861		€ 11,167		€ 9,306								€ 22,334	€ 0	€ 22,334
9	TUD Jeroen Kalkman Matching	€ 3,722	€ 1,667	€ 22,334	€ 10,000	€ 18,611	€ 8,333							€ 44,667	€ 20,000	€ 64,667
11	TUD Lidy Fratila-Apachitei Matching	€ 1,861	€ 698	€ 11,167	€ 4,539	€ 9,306	€ 5,238							€ 22,334	€ 10,476	€ 32,810
12	TUD Amir Zadpoor Matching	€ 4,386	€ 333	€ 26,319	€ 1,917	€ 21,932	€ 1,250							€ 52,637	€ 3,500	€ 56,137
13	TUD Sasa Kenjeres Matching	€ 931	€ 2,541	€ 5,583	€ 31,849	€ 4,653	€ 83,333							€ 11,167	€ 117,724	€ 128,891
14	EUR Hub Zwart Matching	€ 0		€ 9,567		€ 4,784								€ 14,351	€ 0	€ 14,351
15	EUR Hedwig Blommestein Matching	€ 1,913		€ 11,481		€ 9,567								€ 22,962	€ 0	€ 22,962
16	HOUDINI anticipated grants Matching					€ 14,003	€ 7,125	€ 266,123	€ 86,100	€ 266,123	€ 93,845	€ 221,769	€ 93,398	€ 768,018	€ 280,468	€ 1,048,486
	Totale bijdrage onderzoeksorganisaties	€ 53,126	€ 12,906	€ 301,895	€ 94,305	€ 252,586	€ 143,614	€ 310,100	€ 86,100	€ 302,355	€ 93,845	€ 236,769	€ 93,398	€ 1,456,832	€ 524,168	€ 1,981,000
Ondernemingen																
17	BIOND Matching		€ 300		€ 1,800		€ 1,800		€ 1,800		€ 1,800		€ 1,500	€ 0	€ 9,000	€ 9,000
	Totale bijdrage ondernemingen	€ 0	€ 300	€ 0	€ 1,800	€ 0	€ 1,800	€ 0	€ 1,800	€ 0	€ 1,800	€ 0	€ 1,500	€ 0	€ 9,000	€ 9,000
Overige (private) partners																
18	hDMT Matching		€ 333		€ 2,000		€ 2,000		€ 2,000		€ 2,000		€ 1,667	€ 0	€ 10,000	€ 10,000
	Totale overige bijdrage	€ 0	€ 333	€ 0	€ 2,000	€ 0	€ 2,000	€ 0	€ 2,000	€ 0	€ 2,000	€ 0	€ 1,667	€ 0	€ 10,000	€ 10,000
	Convergence Grant HOUDINI	€ 66,667		€ 400,000		€ 400,000		€ 400,000		€ 400,000		€ 333,333		€ 2,000,000		€ 2,000,000
		in cash	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**			**Totaal**
	Totaal matched budget per jaar	€ 53,126	€ 13,539	€ 301,895	€ 98,105	€ 252,586	€ 147,414	€ 310,100	€ 89,900	€ 302,355	€ 97,645	€ 236,769	€ 96,565			€ 4,000,000
																Funding gap
																1
	Totaal Matching budget	€ 66,666		€ 400,000		€ 400,000		€ 400,000		€ 400,000		€ 333,334		€ 2,000,000		

bi/ond. Nourishing, Stimulating and Monitoring Cells

BIOND SOLUTIONS B.V.
Molengraaffsingel 12,
2629 JD Delft
The Netherlands

Subject: *Letter of support for the HOUDINI PROJECT*

Dear Dr. Wytske van Weerden,

The purpose of this letter is to formally communicate that BIOND Solutions B.V. (BI/OND) would be delighted to enable the research project proposal entitled "HOUDINI".
This ambitious project aims to drive forward the routine use of in vitro experimental systems using organ on chip devices, pushing for better models of metastatic disease.

This HOUDINI project and the CONVERGENCE program highly reflect our company mission in many ways. BI/OND strongly believes that inclusive and precise treatments must be integrated into medicine. Moreover, this project represents a unique opportunity to evangelize the new generation of scientists on the importance of cross-fertilization between complementary fields, like engineering and biology. As a spin-off of the Technical University of Delft, we know the importance of translating engineering advancement into the biological and business field.

BIOND Solutions would be delighted to support the project by:

1. providing 300 microfluidic chips for the whole project duration at a discounted price.
2. provide the first 16 hours of assistance for free where we share our proprietary protocol and know-how
3. Two free trainings (one engineering and one biological) at the project start.

The in-kind contribution can be quantified as equal to €9.000.

I would like to wish you success with submitting your proposal and await further information on the project with interest.

Sincerely,
By BIOND Solutions B.V.

Name: Dr Cinzia Silvestri
Function: CEO and founder
Date and Place: 22 January 2021, Delft

InSphero AG
Dr. Olivier Frey
Wagistrasse 27a
CH-8952 Schlieren

Phone +41 44 515 0490
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www.insphero.com
olivier.frey@insphero.com

Dr. W.M. van Weerden
Dept. of Urology
Erasmus University Medical Center
Rotterdam
The Netherlands

Subject: Letter of Support for the **Flagship Program Call 221 proposal 'HOUDINI'**
(Holistic approach to UnderstanD and target the metastatic Niche')

Schlieren, March 14th, 2022

Dear Dr. van Weerden,

I, Olivier Frey, in my position as Vice President of InSphero AG, Switzerland, express my support for the HOUDINI project proposal.

InSphero is the pioneer of industrial-grade, 3D-cell-based assay solutions and scaffold-free 3D organ-on-a-chip technology. InSphero works with partners in academia and industry to advance drug discovery and understanding of healthy and diseased human biology using an inspired combination of bioengineering, microtechnologies, and computational approaches. We create miniature organs and biological systems to better predict the patient response in vitro. InSphero is specialized in delivering assay-ready and custom 3D models derived from liver, pancreatic islet, and tumor tissues, to provide unrivalled biological insight into liver toxicology, metabolic diseases (e.g., diabetes and liver diseases), and oncology.

We are excited to hear about the Flagship Program Call 221 proposal 'HOUDINI' with its biomedical and technology-driven projects on metastatic modelling and essential microenvironmental interactions. We see major overlap with our current interests and activities, and a great potential to expand the field of application of InSphero's technology. As and industrial partner of your project, we will support your research by providing access to our organ-on-chip platform and support the needs of your laboratory with high-quality ready-to-use liver microtissues from our production unit. Their physiological function and modular design, respectively, makes them ideal candidates for the e.g. intended post-mechanical measurements of one of the projects.

Sincerely yours,

Dr. Olivier Frey
Vice President
Head of Technologies and Platforms

HUMAN ORGAN AND DISEASE MODEL TECHNOLOGIES

To: Dr Wytske van Weerden
Dept. of Urology
Erasmus University Medical Center
Rotterdam
The Netherlands

Subject: Letter of Intent for the Flagship Program Call 2021 proposal 'HOUDINI (HOlistic approach to UnderstanD and target the metastatIc Niche'

Eindhoven, 27 February 2022

Dear Dr van Weerden,

With this letter we declare on behalf of the Institute for human organ and Disease Model Technologies (hDMT) our interest in and support for your research proposal 'HOUDINI' (HOlistic approach to UnderstanD and target the metastatIc Niche, submitted in the Flagship Program call 2021.

As hDMT, we have read and have been fully informed about your application and we are very interested in participating in this research project.

The Institute for human organ and Disease Model Technologies (hDMT) is a consortium of renowned scientists with backgrounds in engineering, biology, physics, chemistry, pharmacology and medicine. hDMT's objective is to develop human organ and disease models that closely mimic healthy and diseased human tissues and organs, through Organ-on-Chip technology and make them available to interested users for all kind of applications by open access publication. The affiliated researchers share their knowledge, expertise and facilities to create Organ-on-Chip models, by integrating state-of-the-art human stem cell technologies and top-level engineering with a wide variety of other expertise. Currently the Consortium consists of 14 partners, including academic research institutions, university medical centers and knowledge institutes. hDMT can valorize research results through a growing network of collaborating companies.

hDMT has built a European Organ-on-Chip network of research groups in most European countries and established the European Organ-on-Chip Society (EUROoCS). EUROoCS encourages and develops Organ-on-Chip research and provides opportunities to share and advance knowledge and expertise in the field towards better health for all.

Many of the multidisciplinary researchers within HOUDINI, that are involved in the development of cancer-dissemination models, belong to the partner organizations of hDMT and are already collaborating with other scientists in the Cancer-on-Chip theme group. The proposed research fits well with hDMT's mission and with the current hDMT research priorities. HOUDINI will provide innovative model systems for cancer metastasis, that do not only focus on tumor cell targeting, but are also taking into account the important role of the microenvironment within the metastatic niche. If successful, this project will result in a big step forward towards new ways of early detection of (micro)metastasis and new opportunities to prevention of metastatic disease.

This will contribute to hDMT's dream: Imagine that once there will be Organ-on-Chip models that mimic any part of the human body, to study human sickness and health, for personalized disease treatment or even prevention, available and affordable for everyone.

hDMT will contribute to the project by helping with the organization of scientific meetings, trainings and workshops. This is also in line with the ambitions of hDMT INFRA, a national infrastructure on Organ-on-Chip with Service Centers of Expertise, that aims to bring Organ-on-Chip models closer to application and is currently being set up by hDMT. In addition, hDMT can enable active participation of the HOUDINI project researchers in the meetings, organized by hDMT and EUROoCS, and can help in disseminating the HOUDINI project results to the national, European, and international Organ-on-Chip community of research scientists, clinicians, regulators, patient organizations, and industry leaders.

hDMT will provide an ***in-kind* contribution** as itemised in the table below.

Item or activity	*# of item(s) *(Cost-)price OR hours*rate*	*Total Amount*
Co-organization of scientific meetings, trainings and workshops	50 hrs in 5 yrs (€100/hr)	5,000 euro
Dissemination of project results	50 hrs in 5 yrs (€100/hr)	5,000 euro
Total estimated value		10,000 euro

Yours sincerely,

On behalf of the Executive Board of hDMT,

Dr. A.J.M. (Janny) van den Eijnden-van Raaij
Managing Director
Tel: +31(0)6 12998074
Email: j.vandeneijnden@hdmt.technology
Website: www.hdmt.technology

Prostaatkankerstichting

Erasmus MC
T.a.v. mevrouw dr. ir. Wytske M. van Weerden
Dept. of Urology, BE362
Postbus 2400
3000 CA ROTTERDAM

Utrecht, 7 maart 2022
Betreft: adhesiebetuiging Prostaatkankerstichting

Geachte mevrouw Van Weerden,

U heeft contact opgenomen met de Prostaatkankerstichting (PKS) met het verzoek steun te verlenen aan het onderzoeksvoorstel getiteld 'HOUDINI' (HOlistic approach to UnderstanD and target the metastatIc NIche).

De Prostaatkankerstichting heeft de projectaanvraag besproken binnen haar kwaliteitsgroep en vooral gelet op het belang voor mannen met prostaatkanker. Dit onderzoek beoogt de ontwikkeling van modelsystemen die de fysiologie en complexiteit van menselijke weefsels weergeven.
In combinatie met andere gegevens kan hiermee naar verwachting beter inzicht worden verkregen in de processen die uitzaaiingen van (prostaat)kanker veroorzaken. Uiteindelijk kan dit leiden tot nieuwe therapeutische aangrijpingspunten om ontstaan en groei van uitzaaiingen te verminderen of zelfs te voorkomen. Voor patiënten met prostaatkanker zijn de uitkomsten van dit onderzoek van groot belang omdat daarmee hopelijk de behandelmogelijkheden kunnen worden uitgebreid en verbeterd.

Onderhavig onderzoek vindt de Prostaatkankerstichting vanuit patiëntperspectief van belang, omdat het kan bijdragen aan het verbeteren van de behandeling van patiënten met prostaatkanker. Daarom wil de stichting het betreffende project dat voor subsidie zal worden ingediend graag van een positief advies voorzien.

Mochten er nog vragen aan ons zijn, dan zijn we te allen tijde bereid u of de onderzoekers ten dienste te zijn. Wij zouden graag zien dat u de resultaten na afloop van de studie met de Prostaatkankerstichting wilt delen, zodat wij die via onze website en het kwartaalblad "Nieuws" onder de doelgroep kunnen verspreiden.

Met vriendelijke groet,

C. van den Berg
Voorzitter

Nederlandse Leverpatiënten Vereniging

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Erasmus MC
t.a.v. dr. W. van Weerden
Voorzitter HOUDINI consortium
Postbus 2040
3000 CA Rotterdam

Hoogland 22 februari 2022
Ref: 2022.02.34/JW

Geachte dr. Van Weerden,

Met grote belangstelling heeft de Nederlandse Leverpatienten Vereniging (NLV) kennis genomen van het voornemen van het Erasmus MC een nauw samenwerkingsverband aan te gaan met de Technische Universiteit Delft en de Erasmus Universiteit, gericht op *Health and Technology*. De plannen voor een convergence wetenschapsgemeenschap (Flagship), samen met genoemde instituten verheugen ons zeer. Wij zijn ervan zijn overtuigd dat dit alle processen rondom de zorg voor en van de leverpatiënt zal bespoedigen.

De NLV is al bij vele ontwikkelingen betrokken die betrekking hebben op diagnostiek, behandelinterventies, evenals de ontwikkelingen op het gebied van studies naar primaire en secundaire leverkanker. Wij weten dat er al goede ontwikkelingen gaande zijn op het gebied van leverkanker modellen waarbij organoïden en tumor matrix worden gecombineerd om een beter tumor model te hebben voor onderzoek naar nieuwe therapie. Een samenwerkingsverband zullen onder meer deze en nieuwe ontwikkelingen alleen maar ten goede komen.

Alleen door samenwerking met alle facetten van de wetenschap kan het uiteindelijke doel worden bereikt: het verbeteren van de kwaliteit van leven voor patiënten. Kortom, wij steunen dit fantastische initiatief van harte.

Met vriendelijke groet,
mede namens het bestuur

drs. José A. Willemse
directeur

Erasmus MC
Universitair Medisch Centrum Rotterdam

Urologie
dr. J.L. Boormans
Afdelingshoofd Urologie

Dr W.M. van Weerden
Dept of Urology
Erasmus University Medical Center
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Uw kenmerk HJvA/van Weerden
Datum 10 maart 2022

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3015 CN Rotterdam

Onderwerp: Letter of Support for the Flagship Program Call 221 proposal 'HOUDINI' (Holistic approach to UnderstanD and target the metastatic Niche'

Geachte Dr. van Weerden,

Ik, dr. Joost Boormans, in mijn functie als afdelingshoofd van de afdeling Urologie van het Erasmus Medical Center geef blijk van mijn steun aan het HOUDINI projectvoorstel.

Met vriendelijke groeten,

J.L. Boormans

Dr. Joost L. Boormans
Afdelingshoofd Urologie
Erasmus MC, Rotterdam

IV References

[1] Mattei F, Andreone S, Mencattini A, De Ninno A, Businaro L, Martinelli E, et al. Oncoimmunology Meets Organs-on-Chip. Front Mol Biosci 2021;8.

[2] Kamm RD, Bashir R, Arora N, Dar RD, Gillette MU, Griffith LG, et al. Perspective: The promise of multi-cellular engineered living systems. APL Bioeng 2018;2:040901.

[3] Chakrabarty S, Quiros-Solano WF, Kuijten MMP, Haspels B, Mallya S, Lo CSY, et al. A Microfluidic Cancer-on-Chip Platform Predicts Drug Response Using Organotypic Tumor Slice Culture. Cancer Res 2022;82:510–20.

[4] Mout L, van Dessel LF, Kraan J, de Jong AC, Neves RPL, Erkens-Schulze S, et al. Generating human prostate cancer organoids from leukapheresis enriched circulating tumour cells. Eur J Cancer 2021;150:179–89.

[5] Van Hemelryk A, Mout L, Erkens-Schulze S, French PJ, van Weerden WM, van Royen ME. Modeling Prostate Cancer Treatment Responses in the Organoid Era: 3D Environment Impacts Drug Testing. Biomolecules 2021;11:1572.

[6] Boot RC, Koenderink GH, Boukany PE. Spheroid mechanics and implications for cell invasion. Adv Phys X 2021;6.

[7] Burla F, Tauber J, Dussi S, van der Gucht J, Koenderink GH. Stress management in composite biopolymer networks. Nat Phys 2019;15:549–53.

[8] Burla F, Dussi S, Martinez-Torres C, Tauber J, van der Gucht J, Koenderink GH. Connectivity and plasticity determine collagen network fracture. Proc Natl Acad Sci 2020;117:8326–34.

[9] Hub Zwart N. Genomics and self-knowledge: implications for societal research and debate. New Genet Soc 2007;26:181–202.

[10] Zwart H. From primal scenes to synthetic cells. Elife 2019;8.

[11] Zwart H. Continental Philosophy of Technoscience. vol. 38. Cham: Springer International Publishing; 2022.

[12] van Tienderen GS, Li L, Broutier L, Saito Y, Inacio P, Huch M, et al. Hepatobiliary tumor organoids for personalized medicine: a multicenter view on establishment, limitations, and future directions. Cancer Cell 2022.

[13] Baroncelli M, van der Eerden BC., Chatterji S, Rull Trinidad E, Kan YY, Koedam M, et al. Human Osteoblast-Derived Extracellular Matrix with High Homology to Bone Proteome Is Osteopromotive. Tissue Eng Part A 2018;24:1377–89.

[14] Jadaun PK, Zhang S, Koedam M, Demmers J, Chatterjee S, Leeuwen JP, et al. Inhibition of hypoxia-induced Mucin 1 alters the proteomic composition of human osteoblast-produced extracellular matrix, leading to reduced osteogenic and angiogenic potential. J Cell Physiol 2022;237:1440–54.

[15] Nakauma-González JA, Rijnders M, van Riet J, van der Heijden MS, Voortman J, Cuppen E, et al. Comprehensive Molecular Characterization Reveals Genomic and Transcriptomic Subtypes of Metastatic Urothelial Carcinoma. Eur Urol 2022.

[16] Rijnders M, Balcioglu HE, Robbrecht DGJ, Oostvogels AAM, Wijers R, Aarts MJB, et al. Anti-PD-1 Efficacy in Patients with Metastatic Urothelial Cancer Associates with Intratumoral Juxtaposition of T Helper-Type 1 and CD8+ T cells. Clin Cancer Res 2022;28:215–26.

[17] Mout L, van Dessel LF, Kraan J, de Jong AC, Neves RPL, Erkens-Schulze S, et al. Generating human prostate cancer organoids from leukapheresis enriched circulating tumour cells. Eur J Cancer 2021;150:179–89.

[18] Virakul S, Somparn P, Pisitkun T, van der Spek PJ, Dalm VASH, Paridaens D, et al. Integrative Analysis of Proteomics and DNA Methylation in Orbital Fibroblasts From Graves' Ophthalmopathy. Front Endocrinol (Lausanne) 2020;11:619989.
[19] Ekronarongchai S, Palaga T, Saonanon P, Pruksakorn V, Hirankarn N, van Hagen PM, et al. Histone Deacetylase 4 Controls Extracellular Matrix Production in Orbital Fibroblasts from Graves' Ophthalmopathy Patients. Thyroid 2021;31:1566–76.
[20] van den Bossche WBL, Vincent AJPE, Teodosio C, Koets J, Taha A, Kleijn A, et al. Monocytes carrying GFAP detect glioma, brain metastasis and ischaemic stroke, and predict glioblastoma survival. Brain Commun 2021;3:fcaa215.
[21] Luque-Martin R, Mander PK, Leenen PJM, Winther MPJ. Classic and new mediators for in vitro modelling of human macrophages. J Leukoc Biol 2021;109:549–60.
[22] Stavast CJ, Leenen PJM, Erkeland SJ. The interplay between critical transcription factors and microRNAs in the control of normal and malignant myelopoiesis. Cancer Lett 2018;427:28–37.
[23] van Beek AA, Van den Bossche J, Mastroberardino PG, de Winther MPJ, Leenen PJM. Metabolic Alterations in Aging Macrophages: Ingredients for Inflammaging? Trends Immunol 2019;40:113–27.
[24] Bunnik EM, Timmers M, Bolt IL. Ethical Issues in Research and Development of Epigenome-wide Technologies. Epigenetics Insights 2020;13:251686572091325.
[25] Dupras C, Bunnik EM. Toward a Framework for Assessing Privacy Risks in Multi-Omic Research and Databases. Am J Bioeth 2021;21:46–64.
[26] Bunnik EM, Dondorp WJ, Bredenoord AL, de Wert G, Cornel MC. Mainstreaming informed consent for genomic sequencing: A call for action. Eur J Cancer 2021;148:405–10.
[27] Verhoef EI, van Cappellen WA, Slotman JA, Kremers G-J, Ewing-Graham PC, Houtsmuller AB, et al. Three-dimensional architecture of common benign and precancerous prostate epithelial lesions. Histopathology 2019;74:1036–44.
[28] Verhoef EI, van Cappellen WA, Slotman JA, Kremers G-J, Ewing-Graham PC, Houtsmuller AB, et al. Three-dimensional analysis reveals two major architectural subgroups of prostate cancer growth patterns. Mod Pathol 2019;32:1032–41.
[29] Kolijn K, Verhoef EI, Smid M, Böttcher R, Jenster GW, Debets R, et al. Epithelial-Mesenchymal Transition in Human Prostate Cancer Demonstrates Enhanced Immune Evasion Marked by IDO1 Expression. Cancer Res 2018;78:4671–9.
[30] Verweij FJ, Balaj L, Boulanger CM, Carter DRF, Compeer EB, D'Angelo G, et al. The power of imaging to understand extracellular vesicle biology in vivo. Nat Methods 2021;18:1013–26.
[31] Hartjes TA, Slotman JA, Vredenbregt MS, Dits N, Van der Meel R, Duijvesz D, et al. EVQuant; high-throughput quantification and characterization of extracellular vesicle (sub)populations. BioRxiv 2020:2020.10.21.348375.
[32] Liu Q, Muralidharan A, Saateh A, Ding Z, ten Dijke P, E. BP. A programmable multifunctional 3D cancer cell invasion micro platform. Small 2022.
[33] de Wit J, Angelopoulos K, Kalkman J, Glentis G-O. Fast and accurate spectral-estimation axial super-resolution optical coherence tomography. Opt Express 2021;29:39946.
[34] van Rooij J, Kalkman J. Large-scale high-sensitivity optical diffraction tomography of zebrafish. Biomed Opt Express 2019;10:1782.

[35] Nouri-Goushki M, Angeloni L, Modaresifar K, Minneboo M, Boukany PE, Mirzaali MJ, et al. 3D-Printed Submicron Patterns Reveal the Interrelation between Cell Adhesion, Cell Mechanics, and Osteogenesis. ACS Appl Mater Interfaces 2021;13:33767–81.
[36] Kramer RCLN, Verlinden EJ, Angeloni L, van den Heuvel A, Fratila-Apachitei LE, van der Maarel SM, et al. Multiscale 3D-printing of microfluidic AFM cantilevers. Lab Chip 2020;20:311–9.
[37] Angeloni L, Ganjian M, Nouri-Goushki M, Mirzaali MJ, Hagen CW, Zadpoor AA, et al. Mechanical characterization of nanopillars by atomic force microscopy. Addit Manuf 2021;39:101858.
[38] Nouri-Goushki M, Angeloni L, Modaresifar K, Minneboo M, Boukany PE, Mirzaali MJ, et al. 3D-Printed Submicron Patterns Reveal the Interrelation between Cell Adhesion, Cell Mechanics, and Osteogenesis. ACS Appl Mater Interfaces 2021;13:33767–81.
[39] Pitta Kruize C, Panahkhahi S, Putra NE, Diaz-Payno P, van Osch G, Zadpoor AA, et al. Biomimetic Approaches for the Design and Fabrication of Bone-to-Soft Tissue Interfaces. ACS Biomater Sci Eng 2021.
[40] Ganjian M, Modaresifar K, Rompolas D, Fratila-Apachitei LE, Zadpoor AA. Nanoimprinting for high-throughput replication of geometrically precise pillars in fused silica to regulate cell behavior. Acta Biomater 2022;140:717–29.
[41] Nouri-Goushki M, Isaakidou A, Eijkel BIM, Minneboo M, Liu Q, Boukany PE, et al. 3D printed submicron patterns orchestrate the response of macrophages. Nanoscale 2021;13:14304–15.
[42] Kenjereš S, van der Krieke JP, Li C. Endothelium resolving simulations of wall shear-stress dependent mass transfer of LDL in diseased coronary arteries. Comput Biol Med 2019;114:103453.
[43] Perinajová R, Juffermans JF, Westenberg JJM, van der Palen RLF, van den Boogaard PJ, Lamb HJ, et al. Geometrically induced wall shear stress variability in CFD-MRI coupled simulations of blood flow in the thoracic aortas. Comput Biol Med 2021;133:104385.
[44] Kenjereš S. On Recent Progress in Modelling and Simulations of Multi-scale Transfer of Mass, Momentum and Particles in Bio-medical Applications. Flow, Turbul Combust 2016;96:837–60.
[45] Blommestein HM, Franken MG, van Beurden-Tan CHY, Blijlevens NMA, Huijgens PC, Sonneveld P, et al. Cost-effectiveness of Novel Treatment Sequences for Transplant-Ineligible Patients With Multiple Myeloma. JAMA Netw Open 2021;4:e213497.
[46] Blommestein HM, de Groot S, Aarts MJ, Vemer P, de Vries R, van Abeelen AFM, et al. Cost-effectiveness of obinutuzumab for chronic lymphocytic leukaemia in The Netherlands. Leuk Res 2016;50:37–45.
[47] Blommestein HM, Verelst SGR, de Groot S, Huijgens PC, Sonneveld P, Uyl-de Groot CA. A cost-effectiveness analysis of real-world treatment for elderly patients with multiple myeloma using a full disease model. Eur J Haematol 2016;96:198–208.
[48] Linke F, Aldighieri M, Lourdusamy A, Grabowska AM, Stolnik S, Kerr ID, et al. 3D hydrogels reveal medulloblastoma subgroup differences and identify extracellular matrix subtypes that predict patient outcome. J Pathol 2020:path.5591.
[49] Arlt A, von Bonin F, Rehberg T, Perez-Rubio P, Engelmann JC, Limm K, et al. High CD206 levels in Hodgkin lymphoma-educated macrophages are linked to matrix-remodeling and lymphoma dissemination. Mol Oncol 2019:1878-0261.12616.
[50] Linke F, Harenberg M, Nietert MM, Zaunig S, von Bonin F, Arlt A, et al. Microenvironmental interactions between endothelial and lymphoma cells: a role for the canonical WNT pathway in Hodgkin lymphoma. Leukemia 2016.

[51] Huang P, Kim K, Schermer M. Ethical Issues of Digital Twins for Personalized Health Care Service: Preliminary Mapping Study. J Med Internet Res 2022;24:e33081.

[52] Huang P. COVID-19 vaccination and the right to take risks. J Med Ethics 2021:medethics-2021-107545.

[53] Leeneman B, Franken MG, Coupé VMH, Hendriks MP, Kruit W, Plaisier PW, et al. Stage-specific disease recurrence and survival in localized and regionally advanced cutaneous melanoma. Eur J Surg Oncol 2019;45:825–31.

[54] Leeneman B, Blommestein HM, Coupé VMH, Hendriks MP, Kruit WHJ, Plaisier PW, et al. Real-world healthcare costs of localized and regionally advanced cutaneous melanoma in the Netherlands. Melanoma Res 2021;31:249–57.

[55] Leeneman B, Uyl-de Groot CA, Aarts MJB, van Akkooi ACJ, van den Berkmortel FWPJ, van den Eertwegh AJM, et al. Healthcare Costs of Metastatic Cutaneous Melanoma in the Era of Immunotherapeutic and Targeted Drugs. Cancers (Basel) 2020;12:1003.

[56] Vandekerckhove P, de Mul M, de Groot L, Elzevier HW, Fabels B, Haj Mohammad S, et al. Lessons for Employing Participatory Design When Developing Care for Young People with Cancer: A Qualitative Multiple-Case Study. J Adolesc Young Adult Oncol 2021;10:404–17.

[57] Vandekerckhove P, de Mul M, Bramer WM, de Bont AA. Generative Participatory Design Methodology to Develop Electronic Health Interventions: Systematic Literature Review. J Med Internet Res 2020;22:e13780.

[58] Vandekerckhove P, Vandekerckhove Y, Tavernier R, De Jaegher K, de Mul M. Leveraging User Experience to Improve Video Consultations in a Cardiology Practice During the COVID-19 Pandemic: Initial Insights. J Med Internet Res 2020;22:e19771.